

PLSC 197 / SODA 110N: Social Data, Technology, and Artificial Intelligence

Christopher Zorn

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Basics: Neural Networks

A “neuron”:

$$Y_i = f[\underset{\substack{\uparrow \\ \text{these}}}{a} + \underset{\substack{\uparrow \\ \text{are}}}{w_1} X_{1i} + \underset{\substack{\uparrow \\ \text{parameters}}}{w_2} X_{2i} + \dots \underset{\substack{\uparrow \\ \text{("weights")}}}{w_k} X_{ki}]$$

where:

- Y is an *outcome*
- The X s are *features* (think “predictors”)
- The w s are *weights*
- $f(\cdot)$ is a function that maps the X s to Y
- The weights are *learned* (estimated) from the data

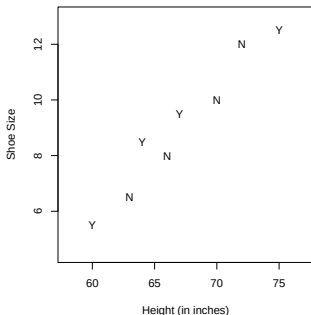
One “Neuron”: Example

Suppose we have:

$$\text{Shoe Size}_i = f[a + w_1(\text{Height})_i + w_2(\text{Weeknd Fan})_i]$$

And some data:

ID	ShoeSize	Height	WeekndFan
1	8.5	64	Yes
2	12.5	75	Yes
3	12.0	72	No
4	6.5	63	No
5	10.0	70	No
6	9.5	67	Yes
7	5.5	60	Yes
8	8.0	66	No



One “Neuron”: Example (continued)

Learned weights (via least squares):

Predictor	Unstandardized w	Standardized w
a (“Intercept”)	-23.9	
Height	0.49	0.98
Sean Penn Fan	0.48	0.11

Predictions:

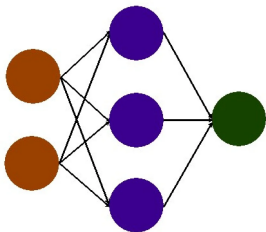
$$\widehat{\text{Shoe Size}}_i = -23.9 + (0.49 \times \text{Height}_i) + (0.48 \times \text{Weeknd Fan}_i)$$

Gives:

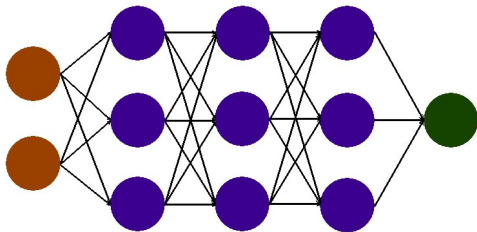
ID	ShoeSize	Height	WeekndFan	Prediction
1	8.5	64	Yes	7.78
2	12.5	75	Yes	13.14
3	12.0	72	No	11.19
4	6.5	63	No	6.81
5	10.0	70	No	10.22
6	9.5	67	Yes	9.24
7	5.5	60	Yes	5.84
8	8.0	66	No	8.27

Neural Network

Artificial Neural Network
(Single Layer ML)



Deep Neural Network
(Multiple Layer ML)



● - Input Layer

● - Hidden Layer

● - Output Layer

Neural Networks and Language

Language occurs *sequentially*:

- Order imparts structure + meaning
- Words' meaning turns on future words:

**“I arrived at the court rested and ready to...
{...play,”
...litigate”}***

- → *recurrent* neural nets (process each word between “court” and {“play”, “litigate”})
- 2017: Transformer models (“self-attention”)...

* Example borrowed from [here](#).

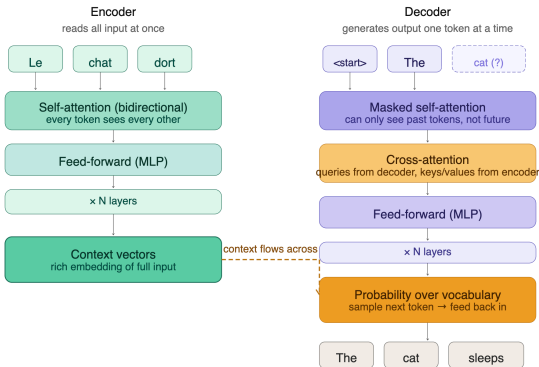
Transformers, Attention, and Language

“Attention”:

- Simultaneously relates every token (word) to every other word,* via *encoders* and *decoders*...
- *Encoders*:
 - Goal: Fullest / most complete picture of language
 - Output: Context-specific word vectors / embeddings
 - Uses: translation, summarization (+ cross-attention)
- *Decoders*:
 - Goal: Predict the next word / token in a sequence
 - Output: Single words in a row
 - Uses: Predictive text, most modern LLMs (GPT, Claude, etc.)

* The math uses dot-products of word embeddings to consider each word in its context; a simple explainer is [here](#).

Transformers, Conceptually



Encoder : Reader :: Decoder : Writer

Large Language Models

“Large Language Models” (LLMs):

- ...are transformer-type models
- ...trained on very large amounts of data [e.g., the Colossal Clean Crawled Corpus (C4): 800 GB of text, 156 billion tokens]
- ...and using decoder-only methods (e.g., generative pre-trained transformer – GPT – circa 2018)
- More recent/sophisticated models include reinforcement learning w/human feedback
- Examples: ChatGPT, Claude, DeepSeek, Gemini, etc.

What Can LLMs Do?

“The basic function of an LLM, namely to generate statistically likely continuations of word sequences, is extraordinarily versatile.”

- **Shanahan (2024)**

LLMs can:

- “Converse”
- Summarize
- Translate
- Generate new content
- Solve puzzles / problems

What Can't / Don't LLMs Do?

LLMs:

- ...don't “know” things (*“knowing that the word ‘Burundi’ is likely to succeed the words ‘The country to the south of Rwanda is...’ is not the same as knowing that Burundi is to the south of Rwanda”*), and
- ...do not have “beliefs” (i.e., can't make truth/falsity judgments / claims with reference to reality). They do, however,
- ...*encode knowledge* (a la an encyclopedia), but they
- ...(in general) do not *reason* – they aren't required to conform to the rules of logic (even when they do).

How Do We Talk (/Think) About LLMs?



an·thro·po·mor·phize

/,anTHrəpə'môr,fiz/

verb

attribute human characteristics or behavior to (a god, animal, or object).

"people's tendency to anthropomorphize their dogs"

Key points:

- Center how LLMs “work” (next-word prediction)
- Use terms like “know” and “believe” only via the *intentional stance* (“...the strategy of interpreting the behavior of an entity...by treating it as if it were a rational agent.”)